Student ID: _____ Signature: _____

PEKI COLLEGE OF EDUCATION
PKCE
LIBRARY
P. O. BOX 14, PEKI, VR.

UNIVERSITY OF GHANA
(**All rights reserved**)
DEPARTMENT OF TEACHER EDUCATION
SCHOOL OF EDUCATION AND LEADERSHIP
COLLEGES OF EDUCATION
END OF SEMESTER ONE EXAMINATIONS FOR LEVEL 300, 2023/2024
B.ED. PROGRAMME

COURSE CODE: TEEG 311
COURSE TITLE: EARLY GRADE SCIENCE

Instruction: Answer **all** questions in Section A and any **three** questions in Section B.
Time: 2 hours

SECTION A
[25 Marks]
Answer all the questions in this section.

STUDENT'S ID NO:_____ SIGNATURE: _____

1. The main reason why plants produce flowers is _____.
   A. To beautify the environment
   B. For medicinal purpose
   C. Sexual reproduction
   D. Dispersal
2. How do flowering plants reproduce?
   A. Seeds
   B. Root
   C. Leaves
   D. None of these
3. Pollination is a process of transferring of pollen grains to _____.
   A. Corolla
   B. Fruit
   C. Flower
   D. Stigma
4. Ovules are present inside _____.
   A. Anther
   B. Calyx
   C. Stigma
   D. Ovary
5. Scattering of seed over large distance is called _____
   A. Dormancy
   B. Dispersal
   C. Pollination
   D. Germination

6. The term used to describe the process by which living things give rise to new ones of their own kind is _____.
   A. Growth
   B. Irritability
   C. Reproduction
   D. Respiration

7. Identify the depiction of sexual reproduction in plants from the following options.
   A. Planting cassava sticks
   B. Planting okro seed
   C. Planting the rhizome of ginger
   D. Planting the corm of cocoyam

8. What is the sexual reproductive organ in a flowering plant?
   A. Flower
   B. Leaf
   C. Root
   D. Stem

STUDENT'S ID NO:_____ SIGNATURE: _____

9. An example of an artificial vegetative propagation method is _____.
   A. Budding
   B. Bulb
   C. Corm
   D. Rhizome

10. What is the name of the plant that reproduces by using the bulb?
   A. Cocoyam
   B. Ginger
   C. Plantain
   D. Shallot

11. What is the term used to describe a short, vertical underground stem that is swollen with stored food?
   A. Bulb
   B. Corm
   C. Rhizome
   D. Sucker

12. Which of the following plants is generally NOT reproduced by the stem tuber?
   A. Potato
   B. Sweet potato
   C. Water yam
   D. Yam

13. What is the term for a type of vegetative propagation in which a dormant bud from one plant is joined with the rootstock of a closely related species?
   A. Budding
   B. Grafting
   C. Layering
   D. Marcotting

14. Which of the following is NOT a part of a dicot seed?
   A. Hilum
   B. Micropyle
   C. Receptacle
   D. Seed coat

15. Water is a _____.
   A. Polar solvent
   B. Non polar solvent
   C. An amphipathic solvent
   D. Non polar uncharged solvent

STUDENT'S ID NO:_____ SIGNATURE: _____

16. The elements of a water molecule are held together by which type of bond _____.
   A. Covalent bond
   B. Double bond
   C. Hydrogen bond
   D. Ionic bond

17. Which of the following substances dissolves in water?
   A. Milk, Sugar, Honey
   B. Turpentine, Lemon juice, Petrol
   C. Kerosine, Salt, Soap
   D. Sugar, Vinegar, salt

18. Which of the following human systems are not interdependent?
   A. Respiratory system and circulatory system
   B. Digestive system and circulatory system
   C. Digestive system and urinary system
   D. Respiratory system and nervous system

19. There can be a change in state when there is reduction in temperature of a substance.
   A. False
   B. True

20. The major factor that causes change in state of substances is
   A. Density
   B. Mass
   C. Temperature
   D. Volume

21 Terrestrial and _____ plants have different habitat requirements.

22. Teaching activities for germination may include _____ and hands-on experiments.

23. Classification of animals can be based on their _____ habits.

24. The human body has various parts with specific _____.

25. Clean water is essential for human health and _____ sustainability.

STUDENT'S ID NO:_____ SIGNATURE: _____

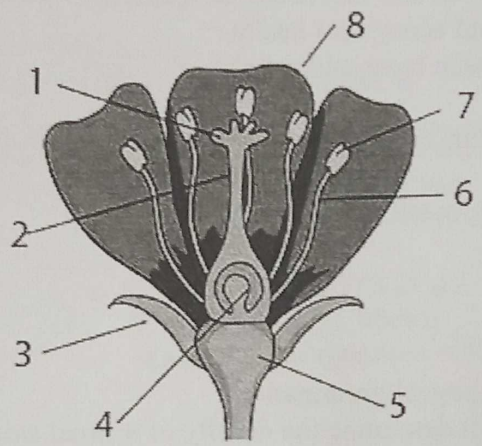
SECTION B

[75 Marks]

Answer any **THREE** questions in this section.

1. a. Give two features of flowers which favour pollination by insects. 2 marks

b. Label the parts numbering from 1 to 8 8 marks



c. Copy and complete the table below.

Characteristics of soil	Sand	Loam	Clay
Water drainage			
Aeration			
Support for plant growth			

(9marks).

d. How will you teach the parts of a **named** flower to early grade learners? (6 marks)

2. a. Describe an experiment to show the components of white light. (5marks)

b. Convert the following temperatures in Kelvin scale to degrees Celsius.

i. 3 K
ii. -20 K
iii. 250 K
iv. 15 K

(8marks)

c. Describe how you will help the following groups of learners in an in early grade inclusive science classroom. (9 marks)

i. Visual learners
ii. Auditory learners
iii. Kinesthetic learners

d. What is an incandescent body? (3 marks)

3. a. Evaluate the effectiveness of **three** child centred teaching methods used to convey the concept of plant propagation to early grade learners. (9 marks)
b. Discuss how soil composition relates to its uses in the environment. (6 marks)
c. Explore the connection between soil characteristics, such as nutrient levels and pH, and their impact on plant growth and ecosystem health. (6 marks)
d. State **three** differences between mass and weight. (4 marks)

4. a. Discuss the impacts of air pollutants on respiratory health, ecosystem functioning, and climate change. (6 marks).
b. State the S.I units for the following quantities.

i. Temperature
ii. Density
iii. Speed
iv. Acceleration (8 marks).

c. Distinguish between epigeal and hypogeal germination. (4 marks).
d. Outline systematically, how you will determine the density of a small stone shaped in the form of a cube. (7 marks).

5. a. Define **Three** qualities of clean water and discuss why **each** quality is important for human health and environmental sustainability. (9 marks)
b. Explain the nature of particles in a substance from a liquid state to the solid state. (4 marks)
c. State the interdependence between **three named** pairs of body systems. (9 marks)
d. State any **three** (3) learner-centered approaches in teaching basic school science. (3 marks)